

Blotted Out

Geoscientist Sankar Chatterjee's theory on the extinction of dinosaurs is gaining wider recognition



1 About 250 million years ago, dinosaurs were the dominant animal species on Earth.

4 The collapse of the food chain and release of noxious gases led to mass extinction, including that of dinosaurs.

2 A meteorite crashed into the Arabian Sea 65 million years ago, creating the world's largest crater.

3 The incredibly powerful impact vapourised rocks, cloaking the planet in dust clouds and blocking the sun.

Seychelles islands off of the Indian tectonic plate, and sent them drifting toward Africa. "The Shiva impact made the western coast of India seismically active," explains Chatterjee. "The Shiva projectile was about 25 miles across," contends Chatterjee. "With a meteorite of this magnitude, it would create a huge crater as soon as it hit the surface. Rocks would be vapourised and dust and debris will be sent into the air that would block out the sun." Red-hot rock would rain from the skies, sparking massive global forest fires and causing acid rain that would kill shelled organisms in the oceans and cause massive collapse of the food chain. Some scientists have speculated that the noxious gases released by the Deccan Traps were crucial factors in the dinosaurs' extinction. "It's very tempting to think that the impact actually triggered the volcanism," says Chatterjee. "But that may not be true. It looks like the volcanism was already happening, and the Shiva impact just made it worse."

What Chatterjee has done through extensive research and extrapolation to count on coincidences

is to put postulates in perspective. "He has, over several years, developed a compelling evidential basis for the idea that the Shiva impact could have created its own tectonic framework. That greater impact appears to have shattered the lithosphere, rifted the Seychelles from India, triggered massive continental flood basalts, such as the Deccan Traps, and produced other far reaching consequences that are vital to the still burgeoning mass extinction debates," says William Glen of the US Geological Survey and an authority on the mass extinction debate.

Chatterjee's team hopes to examine rocks drilled from the centre of the putative crater for clues that would prove that the strange basin was formed by a gigantic impact. "Rocks from the bottom of the crater will give us the telltale sign of the impact event from shattered and melted target rocks. And we want to see if there are breccias, shocked quartz, and an iridium anomaly. Asteroids are rich in iridium, and

such anomalies are thought of as the fingerprint of an impact," says Chatterjee whose interest in paleontology goes back to his days at the Indian Statistical Institute (ISI), Kolkata, after he graduated with a Master's degree in science from Jadavpur University before going to the University of California, Berkeley, in 1976 and then doing a stint with the Smithsonian Institution. His Shiva Crater work is the outcome of a collaboration with Dhiraj Rudra of the ISI.

The geoscientist's calculation may well take the debate on the Earth's dynamic history to a path-breaking new high. "Mine is still a working hypothesis and we need additional data to confirm it," admits Chatterjee. "The work we have done is basically a work in progress. We have to find additional evidence from India to confirm the crater, since it is largely submerged." Like any scientific hypothesis, there are some scientists who will endorse the Shiva hypothesis; others may well oppose it. This is how much science has progressed. ■